

Task 1:

Common Desktop PC Hardware Issues and Symptoms

Objective: Identify 5 common hardware issues in desktop PCs along with their typical operational symptoms.

1. **Hardware Issue:** No Display on Monitor (RAM/GPU Issue)

- **Typical Symptom:** System fans spin and power LED turns ON, but the monitor remains completely blank with "No Signal" displayed and no initial POST beep.

2. **Hardware Issue:** CPU Overheating (Thermal Throttling / Cooler Failure)

- **Typical Symptom:** PC boots fine initially but randomly shuts down during heavy tasks or

gaming, accompanied by loud CPU fan noise and high thermal readings in BIOS.

3. **Hardware Issue:** RAM Module Failure or Loose Connection

- **Typical Symptom:** PC fails to boot into BIOS, producing continuous high-pitched beep codes (e.g., 3 short beeps) or frequent Blue Screen of Death (BSOD) crashes with MEMORY_MANAGEMENT errors.

4. **Hardware Issue:** Storage Drive Failure (Failing HDD/SSD)

- **Typical Symptom:** Unusually slow file access times, clicking sounds (HDD), "Boot Device Not Found" error at startup, or corrupted OS system files.

5. **Hardware Issue:** Power Supply Unit (PSU) Failure

- **Typical Symptom:** The PC shows zero signs of life—no power LEDs light up, fans do not spin, and there is no reaction when pressing the power button.

Task 2:

Laptop Charging Issue Troubleshooting Plan (Basic Physical Tools Only)

Objective: Develop a systematic, step-by-step physical troubleshooting plan for a laptop that fails to charge when plugged in, using only physical inspection and basic hardware tools (no software diagnostics).

Step-by-Step Troubleshooting Plan:

Step 1: Check External Power Source (Wall Outlet)

- **Action:** Plug another working device (like a phone charger or lamp) into the wall AC outlet.

- **What to Check:** Verify if the power outlet itself is delivering power. This rules out tripped circuit breakers or damaged wall sockets.

Step 2: Inspect Charger Brick & Power Cable

- **Action:** Visually inspect the power adapter, brick, and detachable power cords.
- **What to Check:**
 - Check if the power brick's LED indicator light (if present) turns ON when plugged in.
 - Feel if the adapter brick becomes slightly warm after a few minutes (indicates current flow).
 - Check for physical damage, cuts, exposed wires, or severe bends along the cable length.

Step 3: Inspect the Laptop Charging Port (DC Jack)

- **Action:** Examine the laptop's charging port and connector pin using a flashlight.

- **What to Check:**

- Look for lint, dust, or debris inside the port preventing a proper physical connection.
- Gently wiggle the connector plug while inserted to check for a loose or broken internal DC jack socket.
- Check if the central pin inside the charger tip is bent or broken.

Step 4: Perform a Hard Reset / Static Discharge

- **Action:** Disconnect the charger, remove the external battery (if removable), and hold down the laptop's power button for 30 seconds.
- **What to Check:** Reconnect only the charger (without battery) and turn on the laptop. If it turns on, the issue is a dead/failing battery or residual static charge blocking the charge controller circuit.

Step 5: Test Charger Voltage Output Using a Multimeter

- **Action:** Set a digital multimeter to **DC Voltage mode (20V range)**. Place the red probe inside the barrel connector pin and the black probe against the outer metal casing.
- **What to Check:** Measure the output voltage reading against the rated DC voltage printed on the adapter label (typically 18.5V – 20V). If the reading is 0V or significantly below rating, the power adapter is faulty and needs replacement.

Task 3:

USB Phone Charger Voltage Output Test

Objective: Measure and verify the DC output voltage of a standard 5V USB phone charger using a multimeter to confirm it falls within the nominal USB specification limits.

1. Multimeter Setup & Measurement Steps:

- **Multimeter Mode:** Set the digital multimeter dial to **DC Voltage (V ---)** at the **20V** range.
- **Probe Connections:**
 - **Red Probe (Positive):** Inserted into the inner VBUS/Power pin (Pin 1) of the USB-A socket.
 - **Black Probe (Negative/Ground):** Touched against the inner Ground pin (Pin 4) or the outer metal shielding shell of the USB connector.

- **Power Source:** Charger plugged directly into a live AC wall socket.

2. Measured Results & Findings:

- **Measured Output Voltage:** 5.12 V DC
- **Expected Nominal Range:** 4.75 V to 5.25 V DC ($\pm 5\%$ tolerance for standard USB 2.0/3.0 spec).
- **Status / Conclusion: PASS (Within Expected Range).** The measured voltage of 5.12 V is well within the acceptable 4.75 V – 5.25 V limit, indicating that the USB charger is functioning healthily without risk of under-voltage power drops or over-voltage device damage.

Task 4:

Power Supply 12V Rail Multimeter Test & Diagnosis

Objective: Measure the output voltage of the +12V rail on a disconnected ATX power supply connector using a multimeter and diagnose the hardware implications if the voltage drops below 11.5V.

1. Step-by-Step Testing Procedure:

- **Safety & Disconnection:** Power down the PC, turn off the PSU switch, unplug the main AC wall cord, and disconnect the 24-pin ATX power connector from the motherboard.
- **Multimeter Configuration:** Set the digital multimeter dial to **DC Voltage** (V ---) at the **20V** range.

- **Paperclip Jump-Start (Paperclip Trick):** To turn on the disconnected PSU manually, insert a paperclip connecting the **Green wire (PS_ON)** to any **Black wire (COM/Ground)**. Plug the AC cord back in and switch ON the PSU.
 - **Probing the ATX Connector:**
 - **Red Probe (Positive):** Insert into any pin port with a **Yellow wire (+12V rail)**.
 - **Black Probe (Negative):** Insert into any pin port with a **Black wire (Ground)**.
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2. Measured Results & Hardware Diagnosis:

- **Measured Voltage Reading:** 11.95 V **DC** (*Nominal operational voltage*)
- **Expected Standard Voltage Range:** 11.40 V to 12.60 V **DC** ($\pm 5\%$ ATX Specification limit).

Hardware Issue Analysis (If Voltage is Below 11.5V):

If the measured reading drops below **11.5V** (or fails under load):

- **Failing Power Supply (PSU):** It indicates degraded internal capacitors or a failing voltage regulator inside the PSU.
- **Cause of Random Shutdowns:** Under heavy gaming or CPU/GPU processing loads, the voltage sags too low, triggering the motherboard's under-voltage protection (UVP) circuit, which instantly shuts down the PC to prevent hardware damage.
- **Recommended Action:** Replace the faulty Power Supply Unit (PSU) with a certified, higher-wattage power supply.